

Aabu Yousuf Raj

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Professional Profile

My strongest motivation as a researcher is the feeling that every solved problem reveals a deeper question. I am drawn to artificial intelligence, particularly machine learning, deep learning, and computer vision, because it connects rigorous theory with messy real-world patterns that are difficult to measure, interpret, and act upon. I look forward to pursuing research where careful experimentation and model design can produce useful real-world intelligence.

Education

Bachelor of Science (B.Sc.) in Computer Science

BRAC University

CGPA: 3.88/4.00

Sep 2022 to Feb 2026

Transcript

Graduated with Highest Distinction

Notable Courses: Advanced Algorithms, Software Engineering, Artificial Intelligence, Machine Learning, Data Science, Natural Language Processing II, Image Processing, Computer Vision, Neural Networks.

Academic Honors: Merit-Based Scholarship; Dean's List & Vice-Chancellor's List honoree for consistently outstanding academic performance.

Professional Experience

Student Tutor, BRAC University

Teaching Assistance, Student Consultation, Assessment

Feb 2025 to May 2025

Dhaka, Bangladesh

Supported 40+ students through lab-based troubleshooting, one-on-one consultations, Q&A sessions, and targeted guidance for probation students, contributing to stronger understanding of course concepts and improved academic engagement.

Graded assignments, maintained performance records, and organized digital course resources to streamline delivery of course material.

NIPPON AI Dojo Program

Applied AI, AI Engineering, On-the-Job Training

Jun 2026 to Present

Dhaka, Bangladesh

Selected as one of 10 participants from BRAC University from a pool of 1,000+ applicants for an applied AI training program organized by Chowa Giken Corporation and AI Samurai Co. Ltd., with support from METI, Japan.

Participating in industry-oriented on-the-job training focused on real-world AI development and Japanese engineering practices.

Academic Research

A Computer Vision Driven Ecosystem for Cattle Monitoring: Multi-Disease Classification with Severity Grading, Multi-View Individual Identification, and Weight Estimation

Feb 2025 to Present

Accepted for Publication

Undergraduate thesis under the supervision of Dr. Jannatun Noor.

*Accepted for publication in **Computers and Electronics in Agriculture (Elsevier, Q1)** as first author.*

Proposed an RGB-based computer vision ecosystem for precision livestock farming, integrating multi-disease classification with clinically guided severity grading, multi-view individual identification, and reference-object-free body weight estimation using multi-view images and metadata.

Designed and evaluated deep learning pipelines using attention-based hierarchical multi-task learning, YOLOv8-based cattle localization, ConvNeXt-based metric learning for multi-view cattle identification, and ensemble learning over fused RGB image and metadata features.

Achieved 85.88% hierarchical disease-severity accuracy, 89.80% Rank-5 identification accuracy under cross-view domain shift, and 35.10 kg MAE in weight estimation, with Grad-CAM++ based explainability analysis supporting model interpretation for low-cost, low-shot, and non-invasive agricultural intelligence.

Academic Projects

Brain Tumor MRI Classification and Segmentation	<i>Multi-task DL, Medical Imaging, U-Net</i>	Github
Engineered a multi-task deep learning pipeline using U-Net, Attention U-Net, and Re-sUNet variants for simultaneous brain tumor segmentation and classification on MRI datasets. Improved model robustness through transfer learning, augmentation pipelines, and weighted loss functions to address class imbalance.		
Real-Time Traffic Congestion Prediction in Dhaka	<i>CNN, Image Classification, Urban AI</i>	Github
Developed a CNN-based traffic intelligence system to classify congestion severity from street-level images in Dhaka's heterogeneous, non-lane-based urban environment. Collected, labeled, and preprocessed a custom street-imagery dataset to support data-driven traffic analysis and smart-city planning.		
Token-Level NER with RNN Architectures	<i>NLP, Sequence Labeling, BIO Tagging</i>	Github
Conducted a comparative study of RNN, LSTM, GRU, and BiLSTM architectures for token-level Named Entity Recognition using the BIO tagging scheme on news datasets. Evaluated sequence-labeling performance to analyze the role of recurrent and bidirectional context modeling in information extraction.		
Predicting Diseases from Symptom Profiles	<i>Machine Learning, Healthcare AI, Classification</i>	Github
Built a multi-class disease prediction pipeline from patient symptom profiles, applying feature engineering, Chi-Square based feature selection, and comparative classifier evaluation to identify effective models for symptom-driven clinical decision support.		
Multimodal Music Clustering by VAE Variants	<i>Unsupervised Learning, VAE, Multimodal Learning</i>	Github
Built an unsupervised multimodal clustering system using VAE and Conditional VAE variants to group Bengali-English hybrid music in a shared latent space. Combined audio features, lyric embeddings, and genre conditioning, and evaluated cluster structure using visualization and clustering-quality metrics.		

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, Bash, LaTeX
AI & ML Expertise: Computer Vision (CNNs, YOLO, ViT), Large Language Models (GPT, LLaMA), Transformers, RAG, Agentic Systems, Prompt Engineering, Fine-tuning, Custom Model Training
Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face, YOLO
Libraries: NumPy, Pandas, SciPy, Matplotlib, Seaborn, NLTK, Librosa, Albumentations
Tools: Git, GitHub, VS Code, Jupyter, Linux, Docker, CUDA, MySQL, Google Colab, Overleaf, CVAT, Roboflow

Leadership

Vice-President , BRAC University Marketing Association (BUMA)	Oct 2024 to Feb 2026
Led planning, promotion, sponsorship outreach, and logistics for university-wide technical events, seminars, programming contests, career workshops, and Capture The Flag (CTF) competitions, strengthening BUMA's campus and industry presence.	

Test Scores

IELTS Academic: Overall 7.5 (CEFR C1)

References

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